SCSE2233 Software Design & Architecture — Final Exam

Q1(b) Model Answer: Earned Value Management (EVM) (6 marks)

What EVM is (plain words)

EVM checks the health of a project by comparing THREE money numbers: how much work you have **done** (EV), how much work you **planned** to have done by now (PV), and how much you actually **spent** (AC). All other numbers come from these three.

The numbers given in this question

Total tasks	20
Cost per task	$1,000
BAC (total budget)	$20,000
Total time	20 months (1 task each month)
Now	end of month 6
Tasks completed	5
AC (money spent)	$8,000

STEP 1 — get the 3 base numbers

Base number	Put numbers	Result	Why
EV = work done	`(5 / 20) × 20,000`	**`= $5,000`**	5 of 20 tasks finished = 25% of the budget
PV = work planned by now	`(6 / 20) × 20,000`	**`= $6,000`**	by month 6 you planned 6 tasks = 30% of budget
AC = money spent	`given`	**`= $8,000`**	already told in the question

PV is NOT in the equation table — you get it from the schedule: (months passed / total months) × BAC.

STEP 2 — plug into the appendix formulas

	Formula	Put numbers	Result	Meaning
CV	`EV − AC`	`5,000 − 8,000`	**`−$3,000`**	over budget (money)
SV	`EV − PV`	`5,000 − 6,000`	**`−$1,000`**	behind schedule (time)
CPI	`EV / AC`	`5,000 / 8,000`	**`0.63`**	< 1 → over budget
SPI	`EV / PV`	`5,000 / 6,000`	**`0.83`**	< 1 → behind schedule
ETC	`(BAC − EV) / CPI`	`15,000 / 0.625`	**`$24,000`**	cost still needed to finish
EAC	`ETC + AC`	`24,000 + 8,000`	**`$32,000`**	total the project will cost

STEP 3 — write the interpretation (1 sentence each)

CPI = 0.63 → for every $1 spent, only about $0.63 of planned work was earned, so the project is **over budget**.

SPI = 0.83 → the team is completing only about 83% of the planned work pace, so the project is **behind schedule**.

Traps that steal marks

- SV compares against PV (the plan), NOT AC. Never write SV = EV – AC.
- PV comes from MONTHS passed (6/20), not from tasks completed (5/20).
- Always write the formula first, then the numbers — method marks are safe even if arithmetic slips.
- Follow the appendix: EAC = ETC + AC. (A different slide shows EAC = AC + (BAC–EV) = $23,000 — in the exam use the appendix.)